

1 Implementation Guide MDE

This section describes how HYDRA-MDE (Machine Data Collection) interacts with DMC (Dynamic Manufacturing Control). The document is based on the DMC Implementation Guide.

Further applicable documents

The following documents contain basic technical information on DMC or supplement this document:

- DMC Implementation Guide
- DMC SDK (included in DMC delivery)
 - o DMC API documentation
 - o Sample ImplementationGuide

MDE configuration

DMC offers the possibility to integrate MDE (Machine Data Collection). To this end, you must configure three components:

- MachineDataManager
- MachineStatusProvider (three options. Default: No monitoring)
 - o CycleMonitoringMachineStatusProvider (Cyclic monitoring)
 - o ProductionMonitoringMachineStatusProvider (Operating signal monitoring)
- HydraMDEFileWriter

You must set up the MachineDataManager and the MachineStatusProvider as child attribute of the workstation component supporting an MDE functionality.

The document [DMC_MDE.docx](#) describes the configuration settings of this component.

Driver data ensure the correct function of the components *MachineDataManager* and *MachineStatusProvider*. The bridge component provides this driver data. You can find details in the section [2.3.7. Bridge](#). You must configure the bridge component as children attribute. In our reference process, the bridge component is the *OPCDriver*.

The FactoryModel (fm_reference.xml) of the reference process shows how MDE components are used in DMC. Find extracts in the following:

```
<ComponentConfiguration Name="HydraMDEFileWriter"
    Class="mpdv.MachineDataCollector.Dmc.MachineData.HydraMDEFileWriter">
    <Parameter Name="FileTimespan" Value="Minute" />
    <Parameter Name="OutputDirectory" Value="MDE" />
```

```

</ComponentConfiguration>

<ComponentConfiguration Name="PreHeatingMachineDataManager"
    Class="mpdv.MachineDataCollector.Dmc.MachineData.MachineDataManager"
    Children="HydraMDEFileWriter,OpcDriver">
    <Parameter Name="MachineDataManager_CounterConfiguration"
        Value="counter://HeatingCounters.xml" />
    <Parameter Name="MachineDataManager_SendingIntervalToHydra" Value="20" />
</ComponentConfiguration>

<ComponentConfiguration Name="PreHeatingMachineStatusProvider"
    Class="mpdv.MachineDataCollector.Dmc.MachineData.MachineStatusProvider"
    Children="OpcDriver">
    <Parameter Name="MachineStatus_MachineStatusConfiguration"
        Value="wpstatus://HeatingWorkplaceStatusConfiguration.xml" />
    <Parameter Name="MachineStatus_OutputMachineLock" Value="1" />
    <Parameter Name="MachineStatus_OutputMachineDowntime" Value="2" />
</ComponentConfiguration>

<ComponentConfiguration Name="PREHEAT"
    Class="mpdv.MachineDataCollector.Dmc.Entities.Workstation"
    Children="...",
        PreHeatingMachineStatusProvider,
        PreHeatingMachineDataManager
    ,..."
/>

```

Figure 1 MDE processing) shows details of the MDE data collection.

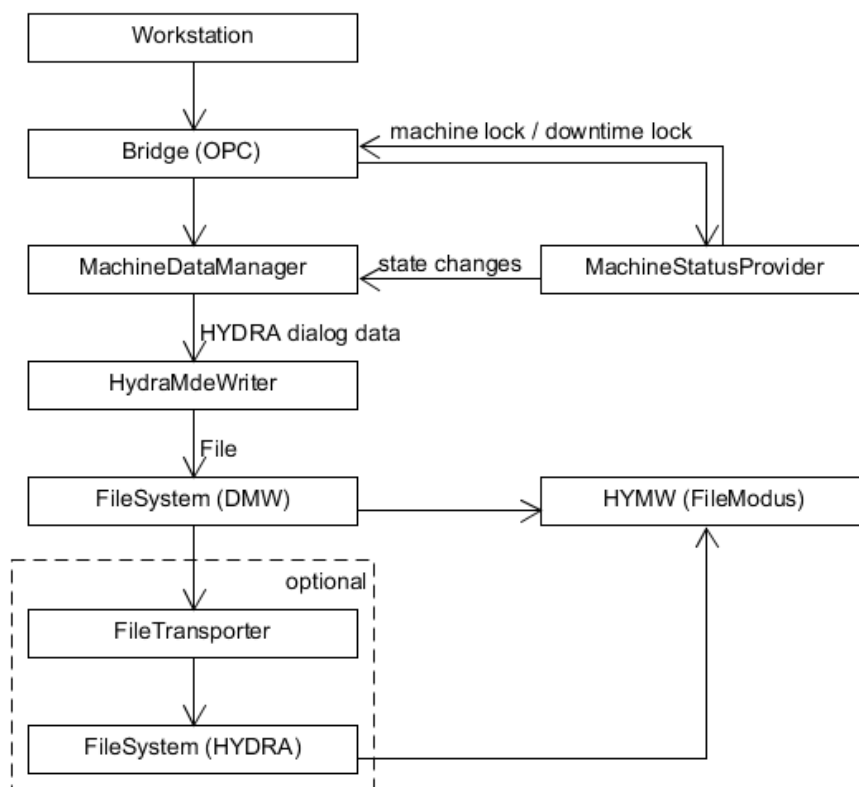


Figure 1 MDE processing

The component *MachineDataManager* sends dialog data to the component *HydraMDEFileWriter* at cyclical intervals or event oriented. The *MachineDataManager* receives status changes of the machine from the *MachineStatusProvider*. With this information, *HydraMDEFileWriter* creates files with Hydra dialog data. The dialog data are stored in the file pipe-separated.

The files created can then be stored directly in a server directory or they can be transferred at a specified interval using the component *FileTransporter*.

If you perform the DMC instance with the MDE components in the server, you do not need the *FileTransporter*. If you use the network to transfer the files, you must use the *FileTransporter*. Using the *FileTransporter*, the transaction is safe. No data is lost during transaction.

```

<ComponentConfiguration Name="HydraMDETransporter"
    Class="mpdv.MachineDataCollector.Core.Components.FileTransporter">
    <Parameter Name="InputDirectory" Value="MDE" />
    <Parameter Name="OutputDirectory" Value="\\DMC-TEST-01\hydra\1\DMC\DMW_01" />
    <Parameter Name="Interval" Value="90" />
    <Parameter Name="SearchPattern" Value="*.mdat" />
</ComponentConfiguration>

```

You must set up a corresponding HYMW service (filemode) for HYDRA to process the data (i.e. load data into the database).

Please refer to the DMC API documentation for further details on the components mentioned.